

Dokumentasi Konfigurasi Host/Hypervisor

Langkah terakhir adalah menyiapkan sisi *host* atau *hypervisor* yang akan dikelola oleh CloudStack:

1. Melakukan pembaruan sistem (*update*) dan pastikan Tailscale juga aktif di sisi host.

```
vboxuser@CloudStack-Host:~$ sudo apt update && sudo apt upgrade -y
```

2. Memastikan Tailscale juga aktif di sisi host.

```
vboxuser@CloudStack-Host:~$ curl -fsSL https://tailscale.com/install.sh | sh
Installing Tailscale for ubuntu noble, using method apt
```

```
+ sudo apt-get install -y tailscale tailscale-archive-keyring
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  tailscale tailscale-archive-keyring
0 upgraded, 2 newly installed, 0 to remove and 3 not upgraded.
Need to get 37.1 MB of archives.
After this operation, 73.0 MB of additional disk space will be used.
Get:1 https://pkgs.tailscale.com/stable/ubuntu noble/main amd64 tailscale amd64 1.98.4 [37.1 MB]
Get:2 https://pkgs.tailscale.com/stable/ubuntu noble/main all tailscale-archive-keyring all 1.35.181 [3,082 B]
Fetched 37.1 MB in 48s (766 kB/s)
Selecting previously unselected package tailscale.
(Reading database ... 85560 files and directories currently installed.)
Preparing to unpack .../tailscale_1.98.4_amd64.deb ...
Unpacking tailscale (1.98.4) ...
Selecting previously unselected package tailscale-archive-keyring.
Preparing to unpack .../tailscale-archive-keyring_1.35.181_all.deb ...
Unpacking tailscale-archive-keyring (1.35.181) ...
Setting up tailscale-archive-keyring (1.35.181) ...
Setting up tailscale (1.98.4) ...
Created symlink /etc/systemd/system/multi-user.target.wants/tailscaled.service → /usr/lib/systemd/system/tailscaled.service.
Scanning processes...
Scanning candidates...
Scanning linux images...

Running kernel seems to be up-to-date.

Restarting services...

Service restarts being deferred:
/etc/needrestart/restart.d/dbus.service
systemctl restart systemd-logind.service
systemctl restart unattended-upgrades.service

No containers need to be restarted.

User sessions running outdated binaries:
vboxuser @ session #1: login[960]
vboxuser @ user manager service: systemd[1206]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
+ [ false = true ]
+ set +x
Installation complete! Log in to start using Tailscale by running:
```

3. tailscale up digunakan untuk mengaktifkan dan menghubungkan perangkat ke jaringan Tailscale VPN.

```
vboxuser@CloudStack-Host:~$ sudo tailscale up

To authenticate, visit:

https://login.tailscale.com/a/e2e17ef016538
```

4. install hypervisor KVM

```
 vboxuser@CloudStack-Host:~$ sudo apt install -y qemu-kvm libvirt-daemon-system libvirt-clients bridge-utils chrony cpu-checker_
```

verifikasi hardware virtualization

```
 vboxuser@CloudStack-Host:~$ kvm-ok
INFO: /dev/kvm exists
KVM acceleration can be used
```

5. add cloudstack key dan repository

```
 vboxuser@CloudStack-Host:~$ sudo mkdir -p /etc/apt/keyrings
vboxuser@CloudStack-Host:~$ wget -O- http://packages.shapeblue.com/release.asc | sudo gpg --dearmor -o /etc/apt/keyrings/cloudstack.gpg
--2026-06-03 12:54:03-- http://packages.shapeblue.com/release.asc
Resolving packages.shapeblue.com (packages.shapeblue.com)... 3.5.69.58, 3.5.72.84, 52.92.32.172, ...
Connecting to packages.shapeblue.com (packages.shapeblue.com)|3.5.69.58|:80... connected.
HTTP request sent, awaiting response... 200 OK
Length: 1778 (1.7K) [application/x-www-form-urlencoded]
Saving to: #STDOUT#

      100%[=====] 1.74K --.-KB/s in 0s
2026-06-03 12:54:04 (97.8 MB/s) - written to stdout [1778/1778]

vboxuser@CloudStack-Host:~$ echo "deb [signed-by=/etc/apt/keyrings/cloudstack.gpg] http://packages.shapeblue.com/cloudstack/upstream/debian/4.22 /" | sudo tee /
etc/apt/sources.list.d/cloudstack.list
deb [signed-by=/etc/apt/keyrings/cloudstack.gpg] http://packages.shapeblue.com/cloudstack/upstream/debian/4.22 /
```

6. install cloudstack agent

```
 vboxuser@CloudStack-Host:~$ sudo apt install cloudstack-agent -y
```

7. Melakukan konfigurasi pada **QEMU Console** dan buat **Virtual Bridge** (jembatan jaringan virtual) agar mesin virtual (VM) nantinya bisa saling berkomunikasi.

```
 vboxuser@CloudStack-Host:~$ sudo nano /etc/libvirt/qemu.conf
```

```
# UNC is configured to listen on 127.0.0.1 by default.
# To make it listen on all public interfaces, uncomment
# this next option.
#
# NB, strong recommendation to enable TLS + x509 certificate
# verification when allowing public access
#
unc_listen = "0.0.0.0"
```

```
 vboxuser@CloudStack-Host:~$ sudo systemctl restart libvirtd
```

Virtual Bridge

```
 vboxuser@CloudStack-Host:~$ sudo nano /etc/netplan/50-cloud-init.yaml
```

```
GNU nano 7.2
network:
  version: 2
  ethernet:
    enp0s3:
      dhcp4: false
      dhcp6: false
  bridges:
    cloudbri0:
      interfaces: [enp0s3]
      dhcp4: true
      macaddress: 08:00:27:26:0a:2a
```

```
vboxuser@CloudStack-Host:~$ sudo netplan apply
```

```
vboxuser@CloudStack-Host:~$ sudo ip link add name cloudbri1 type bridge
[sudo] password for vboxuser:
vboxuser@CloudStack-Host:~$ sudo ip add add 172.16.50.1/24 dev cloudbri1
vboxuser@CloudStack-Host:~$ sudo ip link set cloudbri1 up
```

```
vboxuser@CloudStack-Host:~$ sudo sysctl -w net.ipv4.ip_forward=1
net.ipv4.ip forward = 1
```

```
vboxuser@CloudStack-Host:~$ sudo iptables -t nat -A POSTROUTING -s 172.16.50.0/24 -o enp0s3 -j MASQUERADE
```

Dokumentasi Storage

Pada WSL2/Ubuntu	
Image	Penjelasan
<pre>fauzan@LAPTOP-CIACUEPU:~\$ sudo apt update sudo apt install nfs-ganesha nfs-ganesha-vfs -y</pre>	Setup NFS
<pre>fauzan@LAPTOP-CIACUEPU:~\$ sudo mkdir -p /export/primary sudo mkdir -p /export/secondary sudo chmod 777 /export/primary /export/secondary</pre>	Membuat directory primary untuk VM dan secondary untuk ISO
<pre>fauzan@LAPTOP-CIACUEPU:~\$ sudo nano /etc/ganesha/ganesha.conf</pre>	mengedit isi dari file configuration ganesha
<pre>EXPORT { Export_Id = 1; Path = /export/primary; Pseudo = /export/primary; Access_Type = RW; Squash = No_Root_Squash; SecType = "sys"; Filesystem_Engine = VFS; CLIENT {</pre>	pengaturan pada layanan NFS-Ganesha untuk mendaftarkan dan membagikan dua folder fisik di dalam WSL Ubuntu kamu (/export/primary dan /export/secondary) dengan hak akses baca-tulis penuh (Access_Type = RW), di mana pengaturan No_Root_Squash sengaja diaktifkan agar sistem Apache CloudStack milik temanmu memiliki wewenang tertinggi (root) saat

<pre> Clients = *; } } EXPORT { Export_Id = 2; Path = /export/secondary; Pseudo = /export/secondary; Access_Type = RW; Squash = No_Root_Squash; SecType = "sys"; Filesystem_Engine = VFS; CLIENT { Clients = *; } } </pre>	memodifikasi file, serta menggunakan tanda bintang (*) pada parameter Clients guna mengizinkan seluruh anggota tim yang berada di dalam satu jaringan Tailscale untuk mengakses storage
<pre> fauzan@LAPTOP-CIACUEPU:~\$ ip addr show eth0 grep inet inet 172.19.89.243/20 brd 172.19.95.255 scope global eth0 inet6 fe80::215:5dff:fe35:dffd/64 scope link </pre>	digunakan untuk menyaring dan menampilkan alamat IP internal yang sedang aktif pada kartu jaringan virtual WSL
<pre> fauzan@LAPTOP-CIACUEPU:~\$ sudo systemctl start nfs-ganesha </pre>	
start nfs (running)	
<pre> ● nfs-ganesha.service - NFS-Ganesha file server Loaded: loaded (/usr/lib/systemd/system/nfs-ganesha.service; enabled; preset: enabled) Active: active (running) since Thu 2026-06-04 00:43:10 WIB; 14ms ago Docs: http://github.com/nfs-ganesha/nfs-ganesha/wiki Process: 4270 ExecStart=/bin/bash -c \${NUMACTL} \${NUMAOPTS} /usr/bin/ganesha.nfsd -C \${OPTIONS} \${EPOCH} \${GNODEID} Main PID: 4271 (ganesha.nfsd) Tasks: 4 (limit: 9339) Memory: 1.5M (peak: 2.2M) CPU: 12ms CGroup: /system.slice/nfs-ganesha.service └─4271 /usr/bin/ganesha.nfsd -C -L /var/log/ganesha/ganesha.log -f /etc/ganesha/ganesha.conf -N NIV_EVENT </pre>	
nfs-ganesha status (running)	
<pre> fauzan@LAPTOP-CIACUEPU:~\$ curl -fsSL https://tailscale.com/install.sh sh </pre>	
install tailscale	
<pre> fauzan@LAPTOP-CIACUEPU:~\$ sudo tailscale up --authkey=tskey-auth-kACnnUMDi11CNTRL-AbEGnbbwP8A9zWiaJfck7AYXXmbLvBe9e --accept-routes </pre>	
menambahkan authkey untuk terhubung ke tailscale hendhy	
<pre> fauzan@LAPTOP-CIACUEPU:~\$ tailscale ip -4 100.125.91.118 fauzan@LAPTOP-CIACUEPU:~\$ ip addr show eth0 grep inet awk '{print \$2}' cut -d/ -f1 172.19.89.243 fe80::215:5dff:fe35:dffd </pre>	
storage ping ke tailscale dan show address pada storage	

```
fauzan@LAPTOP-CIACUEPU:~$ sudo systemctl restart nfs-ganesha  
sudo systemctl status nfs-ganesha
```

restart nfs

Windows Powershell

```
PS C:\Windows\system32> netsh interface portproxy add v4tov4 listenport=2049 listenaddress=100.125.91.118 connectport=2049 connectaddress=172.19.89.243  
PS C:\Windows\system32> netsh interface portproxy show all  
  
Listen on ipv4:          Connect to ipv4:  
-----  
Address      Port      Address      Port  
-----  
100.125.91.118  2049      172.19.89.243  2049
```

bertugas untuk meneruskan (forward) trafik data NFS dari jaringan luar (Tailscale) agar bisa masuk ke dalam server NFS yang berjalan di dalam WSL 2.